

ALU with Built-In Self-Test (BIST), Power Gating, and Dynamic Voltage & Frequency Scaling (DVFS) Optimization

Course: CMPE 630 – Digital IC Design

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Lab Section: 01

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Overall Block Diagram

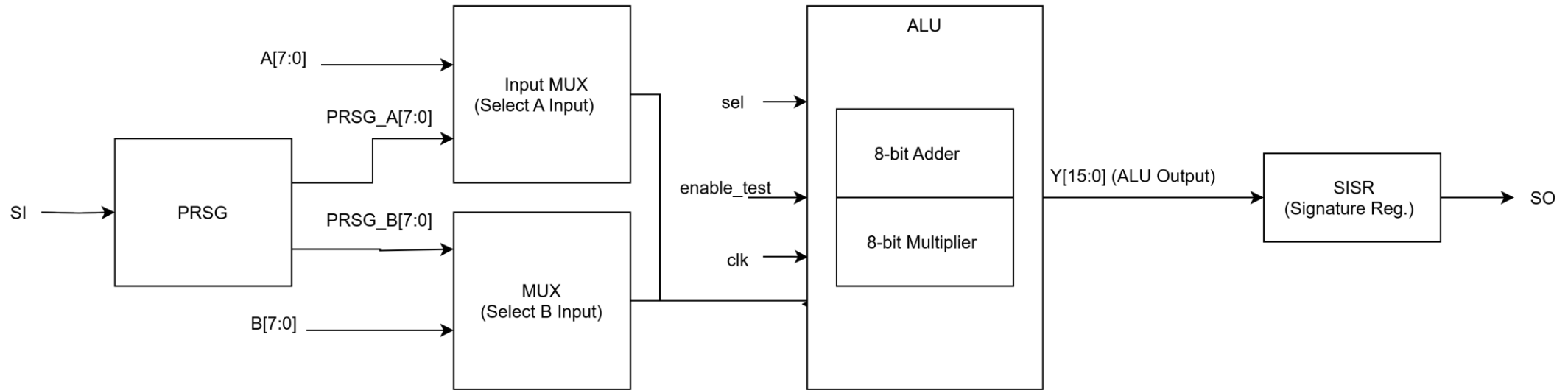


Figure 1: Block diagram of the ALU with Built-In Self-Test (BIST) integration. The Pseudo-Random Sequence Generator (PRSG) supplies randomized test inputs through 2-to-1 multiplexers, allowing selection between normal operation and test mode. The ALU performs either addition or multiplication based on the control signal sel . The result is compressed by the Single Input Signature Register (SISR) into a compact signature, which is scanned out via SO for verification.

Block Diagram Legend

Symbol/Label	Description
A[7:0]	8-bit external input A to ALU
B[7:0]	8-bit external input B to ALU
PRSG	Pseudo-Random Sequence Generator; generates 8-bit test values for A and B
SI	Serial Input (Scan In) to load seed into PRSG during test mode
MUX (A/B)	2-to-1 MUXes select between normal inputs and PRSG outputs
enable_test	Control signal: 1 = use PRSG inputs and enable SISR
sel	Control signal: 0 = Adder, 1 = Multiplier in ALU
clk	Clock signal for synchronous elements (PRSG, SISR, ALU if sequential)
ALU	Arithmetic Logic Unit – executes either $A + B$ or $A \times B$
8-bit Adder	Ripple-carry or carry-lookahead adder using standard cells
8-bit Multiplier	Array or shift-add style multiplier using logic gates
Y[15:0]	16-bit output from ALU (for mult and add)
SISR	Single Input Signature Register – compresses output Y to a small signature
SO	Scan Out – the final compressed signature for comparison to golden sig

Major Component – ALU (Adder + Multiplier)

- Adder: 8-bit ripple-carry adder from Lab 8 using standard cells (AND, XOR, INV)
- Multiplier: New 8-bit unsigned array multiplier built using AND and adder stages
- MUX: Chooses between Adder and Multiplier output using sel bit
- Output: Y[15:0]

Major Component – PRSG + MUX

- PRSG (Pseudo-Random Sequence Generator):
 - 9-bit Galois LFSR
 - Used to generate randomized test inputs

- Input MUXes:
 - Two 2-to-1 MUXes select between external A/B and PRSG outputs
 - Controlled by enable_test signal

Major Component – SISR

- SISR (Single Input Signature Register):
 - 6-bit register with feedback XOR logic
 - Shifts in 1-bit responses from ALU output (e.g., parity or bitwise XOR of Y)
 - Final value is compared to golden signature for pass/fail validation

Optimization – Power Gating and DVFS

- Goal: Save power and improve energy efficiency using multiple strategies
- Power Gating Method:
 - Use sleep transistors to cut VDD to inactive blocks (e.g., Multiplier, PRSG, SISR)
 - Control logic disables power to idle blocks to reduce leakage
- DVFS Method:
 - Adjust operating voltage and frequency dynamically based on workload
 - Lower frequency/voltage for test mode; higher for performance-critical operations
- Why Combine Power Gating and DVFS:
 - Power gating reduces static/leakage power during idle time
 - DVFS reduces dynamic power based on activity level
 - Together, these techniques allow fine-grained control over power and performance tradeoffs, helping to meet timing and energy constraints across different operating modes
 - This combination aligns with ongoing research into adaptive low-power design, and provides a foundation for exploring DVFS schemes in dissertation-level investigations

THANK YOU